

Module-02

Software Development Life Cycle (SDLC) in the Age of AI

Q1. Draw the SDLC lifecycle diagram and briefly explain the workflow.

Q2. Practical AI Exercise

Choose one AI tool (ChatGPT, Claude, GitHub Copilot, Cursor AI, or Gemini) and perform the following:

Generate five user stories for an E-commerce application.

Convert one user story into at least five acceptance criteria.

Ask the AI to create a Sprint Backlog.

Review the AI-generated output and improve it manually.

Explain:

- What the AI did well.
- What corrections did you make?
- Why human review is still necessary.
- Attach screenshots of your AI interaction.

Answer to the question no: 1

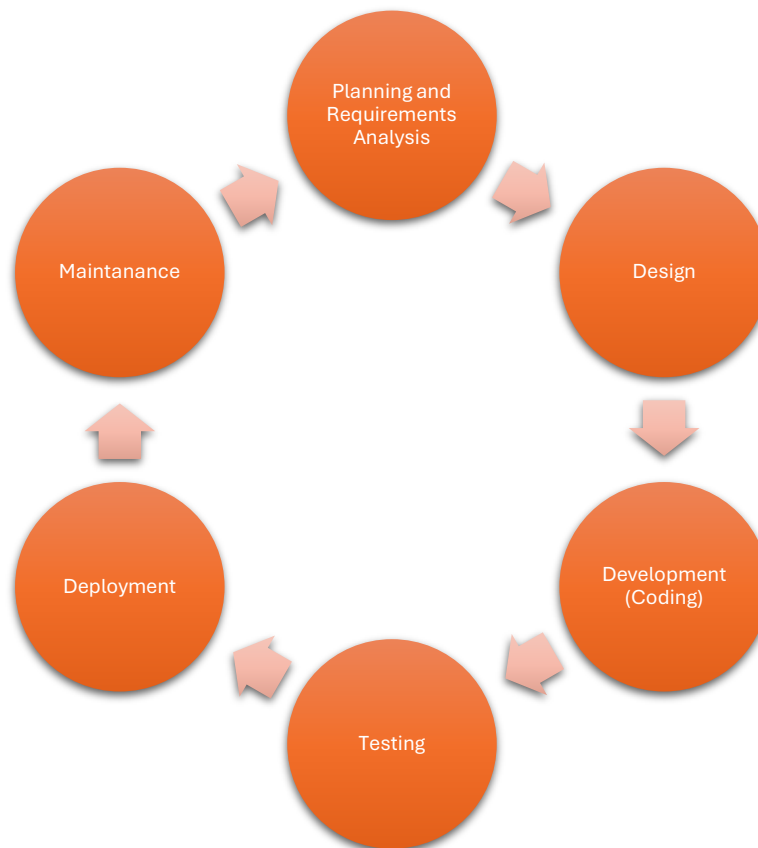


Figure 1: Software Development Life Cycle (SDLC)

1. **Planning and Requirements Analysis:** In this phase business requirements are collected from clients, stakeholders and users. Then business analyst will understand client needs. After that Development team will figure out software requirements. Then team will discuss the features and limitations. Then they'll inform clients and prepare a requirements document like SRS. This phase's goal is to ensure everyone can understand what needs to be done.
2. **Design:** The development team will design database structures, UI/UX design, decides which system architectures will they follow and will prepare High-Level Design (HLD) and Low-Level Design (LLD). This phase will insure how the system will work technically.
3. **Development (Coding):** It's a coding phase where developers will write source codes, create modules and functionalities and perform unit testing. This phase's goal is to convert requirements into a working software product.
4. **Testing:** The developed software will undergo testing to detect bugs and ensure quality. Usually, software testers perform in this phase. This testing includes manual testing, automation testing, functional testing, regression testing, performance testing, bug reporting and verification.
5. **Deployment:** In this phase the software will release to the production environment for users. Necessary software will be installed in servers, configuring environment and releasing the software to users are include in this phase. This phase's goal is to avail the software to all or appropriate users.
6. **Maintenance:** After releasing the software, it needs continuous maintenance and support. Usually this phase handles defects, improve performance, adding new features and security updates. This phase's main goal is to keep the software stable, secure and updated.

Answer to the question no: 2

#	User Story	Acceptance Criteria
1	As a customer, I want to create an account so that I can securely manage my orders and personal information.	1. User can register with name, email, phone number and password.2. Email and phone number must be unique.3. Password must meet security requirements.4. User receives confirmation after successful registration. 5. The system should validate the data provided by the user and ensure that the email address and phone number are not already registered and that the password meets the required strength criteria.
2	As a customer, I want to search and filter products so that I can quickly find the products I need.	1. User can search by product name or keyword.2. User can filter by category, price, brand, etc.3. Search results display matching products.4. A suitable message is shown when no products are found. 5. Search will reset if user reset filter or performing an empty search. 6. User can sort products by newest or price low to high or high to low.
3	As a customer, I want to add products to my shopping cart so that I can purchase multiple products together.	1. User can add an available product to the cart.2. User can increase or decrease quantity.3. Cart displays product name, price, quantity, and subtotal.4. Cart total updates when quantity changes.5. User cannot add more items than available stock. 6. User can reduce the quantity to 0 to remove the product from the cart. 7. User can logout and login later to see the cart again.
4	As a customer, I want to place an order and make a payment so that I can purchase the products in my cart.	1. User can enter/select a delivery address.2. Order summary displays products, quantities, delivery charge, coupon discount and total price.3. User can select an available payment method.4. Successful payment creates an order.5. User receives an order confirmation. 5. System should show proper message if payment is unsuccessful. 6. The system should verify that the product is in stock before processing the payment.
5	As a customer, I want to track my order so that I know the current status of my delivery.	1. User can view their order history.2. Each order displays its current status.3. Status can progress through stages such as Confirmed → Processing → Shipped → Delivered.4. User can view the expected delivery information when available.5. user can give review of his/her older order.

Let's take the "Add Product to Shopping Cart" user story and expand it into at least five acceptance criteria.

User Story

As a customer, I want to add products to my shopping cart so that I can purchase multiple products together.

Acceptance Criteria

1. **Add an available product:** Given a product is in stock, when the customer clicks "Add to Cart", then the product should be added to the shopping cart.
2. **Update cart quantity:** Given a product is already in the cart, when the customer increases or decreases the quantity, then the cart quantity should be updated accordingly. User can reduce the quantity to 0 to remove the product.

3. **Update total price:** When the quantity of a product changes, then the product subtotal and overall cart total should be recalculated correctly.
4. **Prevent adding unavailable products:** Given a product is out of stock, when the customer tries to add it to the cart, then the system should prevent the addition and display an appropriate message.
5. **Prevent exceeding available stock:** Given a product has limited stock, when the customer tries to add more units than are available, then the system should prevent the quantity from exceeding the available stock.
6. **Display cart information:** When the customer opens the shopping cart, then the system should display the product name, price, quantity, subtotal, and total cart amount.
7. **Cart view after re-login:** When the customer log out after adding some products in the cart than after few days he logs in, the cart should be still there with updated stock and price.

Sprint Backlog — E-commerce Application

Sprint Goal: Build the basic customer shopping flow, from registration to adding products to the cart.

ID	User Story / Task	Priority	Story Points	Status
US-01	- Customer registration and account creation - Write & execute tests	High	5	To Do
US-02	- Search and filter products - Write & execute tests	High	5	To Do
US-03	- Shopping cart management (Includes add/remove items, quantity adjustments, stock validation, & total calculation) - Write & execute tests	High	8	To Do

Sprint Summary

- **Sprint Duration:** 2 weeks
- **Sprint Goal:** Implement and test the basic shopping/cart functionality.
- **Total Story Points: 18**
- **Priority:** High-priority customer-facing functionality first.
- **Definition of Done:** Development completed, acceptance criteria passed, test cases executed, and no critical defects remaining.

My Review:

1. What the AI did well

The AI-generated output provided a good starting point:

- The **user stories followed the standard Agile format**: *“As a customer, I want..., so that...”*.
- The stories covered important e-commerce functionality such as **registration, product search, shopping cart, payment, and order tracking**.
- The acceptance criteria were generally **clear and testable**. For example, checking that a product cannot be added when it is out of stock gives QA a concrete condition to verify.
- The Sprint Backlog included useful fields such as **priority, story points, and status**.
- The AI also broke the shopping-cart story into smaller development and testing tasks, which makes the work easier to estimate and track.

2. What corrections did I make?

AI did exceptionally well to write user stories, create acceptance criteria and sprint run. However, as AI is global, it ignores local issues. Such as in User Story 1, while registration, AI exempt phone number. Phone number is essential as delivery partners will contact customers via phone. Also, AI missed coupon discount field. It's also a lucrative option for customers in Bangladesh. Also added that system should check if the user is trying to create the account with already registered email or phone number.

For user story 2, AI didn't tell what will happen if user click filter reset button or search empty. Also, it didn't consider sorting as well. I added both.

I added User can reduce the quantity to 0 to remove the product from the cart, user can logout and login later to see the cart again in user story 3; system should show proper message if payment is unsuccessful and the system should verify that the product is in stock before processing the payment in user story 4 and user can give review of his/her older order in user story 5.

In user story acceptance criteria, I added User can reduce the quantity to 0 to remove the product in serial no 2 and replace no 7 with Cart view after re-login.

In sprint, AI created all cart related work in separate user story. I removed US-04 through US-08 because updating quantities, removing items, calculating totals, checking stock, and displaying cart information are not standalone user stories, but rather core Acceptance Criteria of the main Shopping Cart (US-03) feature. Treating them as separate stories falsely inflates the sprint's story point. Compressing these criteria under US-03 adheres to standard Agile practices, preventing duplicate effort and keeping the sprint backlog clean and accurate. Also, I removed Tests as testing is part of the Definition of Done (DoD) for every story, not a separate User Story on its own. So added Write & execute tests in every user story.

3. Why human review is necessary?

While AI handles basic tasks like drafting user stories, acceptance criteria, and sprint backlogs well, it often misses crucial context. AI lacks domain knowledge about local system requirements and frequently misjudges task complexity—either overcomplicating simple tasks or oversimplifying complex ones. Human review ensures the process remains accurate, context-aware, and fully optimized.

4. Attached Screenshot:

Generate five user stories for an E-commerce application.

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Here are five user stories for an **E-commerce application**, written in the standard Agile format:

#	User Story	Acceptance Criteria
1	As a customer, I want to create an account so that I can securely manage my orders and personal information.	1. User can register with name, email, and password. 2. Email must be unique. 3. Password must meet security requirements. 4. User receives confirmation after successful registration.
2	As a customer, I want to search and filter products so that I can quickly find the products I need.	1. User can search by product name or keyword. 2. User can filter by category, price, brand, etc. 3. Search results display matching products. 4. A suitable message is shown when no products are found.
3	As a customer, I want to add products to my shopping cart so that I can purchase multiple products together.	1. User can add an available product to the cart. 2. User can increase or decrease quantity. 3. Cart displays product name, price, quantity, and subtotal. 4. Cart total updates when quantity changes. 5. User cannot add more items than available stock.
4	As a customer, I want to place an order and make a payment so that I can purchase the products in my cart.	1. User can enter/select a delivery address. 2. Order summary displays products, quantities, delivery charge, and total price. 3. User can select an available payment method. 4. Successful payment creates an order.
		5. User receives an order confirmation.
5	As a customer, I want to track my order so that I know the current status of my delivery.	1. User can view their order history. 2. Each order displays its current status. 3. Status can progress through stages such as Confirmed → Processing → Shipped → Delivered . 4. User can view the expected delivery information when available.

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Simple User Story Template

As a [user/role] ,
I want [feature/action] ,
so that [benefit/reason] .

For example:

As a customer, I want to add products to my shopping cart so that I can purchase multiple products together.

Let's take the "Add Product to Shopping Cart" user story and expand it into at least five acceptance criteria.

User Story

As a customer, I want to add products to my shopping cart so that I can purchase multiple products together.

Acceptance Criteria

1. Add an available product

- Given a product is in stock, when the customer clicks "Add to Cart", then the product should be added to the shopping cart.

2. Update cart quantity

- Given a product is already in the cart, when the customer increases or decreases the quantity, then the cart quantity should be updated accordingly.

3. Update total price

- When the quantity of a product changes, then the product subtotal and overall cart total should be recalculated correctly.

4. Prevent adding unavailable products

- Given a product is out of stock, when the customer tries to add it to the cart, then the system should prevent the addition and display an appropriate message.

5. Prevent exceeding available stock

- Given a product has limited stock, when the customer tries to add more units than are available,

+ Ask anything



Think



- When the customer opens the shopping cart, then the system should display the product name,

Sure. Based on the **E-commerce application** and the user stories we created, here is a sample **Sprint Backlog for Sprint 1**.

Sprint Backlog — E-commerce Application

Sprint Goal: Build the basic customer shopping flow, from registration to adding products to the cart.

ID	User Story / Task	Priority	Story Points	Status
US-01	Customer registration and account creation	High	5	To Do
US-02	Search and filter products	High	5	To Do
US-03	Add products to shopping cart	High	8	To Do
US-04	Update product quantity in cart	High	3	To Do
US-05	Remove products from cart	Medium	2	To Do
US-06	Calculate cart subtotal and total	High	3	To Do
US-07	Validate product stock before adding to cart	High	3	To Do

+

Ask anything

Think

5. Prevent exceeding available stock

- Given a product has limited stock, when the customer tries to add more units than are available, then the system should prevent the quantity from exceeding the available stock.

6. Display cart information

- When the customer opens the shopping cart, then the system should display the product name, price, quantity, subtotal, and total cart amount.

7. Remove product from cart

- Given a product exists in the cart, when the customer clicks "Remove", then the product should be removed and the cart total should be updated.



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US-08	Display cart information	Medium	2	To Do
TEST-01	Write test cases for registration	High	2	To Do
TEST-02	Write test cases for product search/filter	High	2	To Do
TEST-03	Write test cases for shopping cart	High	3	To Do

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Sprint Summary

- Sprint Duration:** 2 weeks
- Sprint Goal:** Implement and test the basic shopping/cart functionality.
- Total Story Points:** 40
- Priority:** High-priority customer-facing functionality first.
- Definition of Done:** Development completed, acceptance criteria passed, test cases executed, and no critical defects remaining.

Example Task Breakdown

For US-03 — Add products to shopping cart:

- Design **Add to Cart** functionality
- Implement Add to Cart API/function
- Connect product page with cart
- Validate product availability
- Update cart item count
- Write test cases
- Execute functional testing
- Fix and retest defects

